Buggy Bot - Autonomus Vehicle

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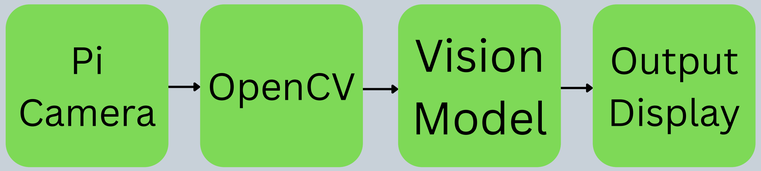
Introduction

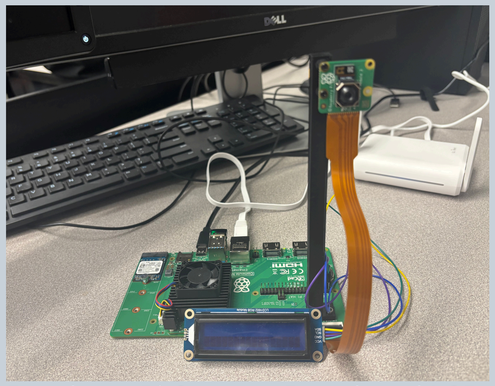
- Autonomous vehicles utilize computer vision, machine learning, and embedded systems to interpret their surroundings and make driving decisions. Accurate traffic sign recognition is essential for autonomous navigation, allowing vehicles to identify road regulations and respond appropriately.
- This project aims to develop a small-scale autonomous car capable of recognizing traffic signs, navigating roadways, and responding to traffic conditions.
- Semester 1 focused on developing a traffic sign recognition system using a Raspberry Pi, camera, OpenCV, and a trained machine learning model.
- The system processes live camera footage to detect and classify traffic signs, displaying the identified sign and its confidence level.
- The primary objective was to establish a functional, real-time vision system that can be integrated with the vehicle's navigation and motor control systems during Semester 2.

Methodology

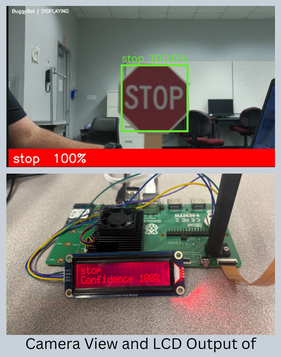
- Vision Model
  - TensorFlow based machine learning model
  - Trained on 3 different traffic sign data sets
  - Classifies detected signs and outputs a predicted label and confidence score
- OpenCV
  - Open-source computer vision library used for real-time image processing.
  - Uses bounding boxes to highlight detected signs and extract images for classification by the vision model
- Custom PCB
  - Designed using KiCad electronic design automation (EDA) software
  - Designed to simplify wiring and support future vehicle assembly and hardware integration.

Processing Pipeline

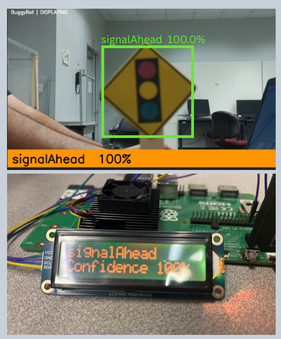




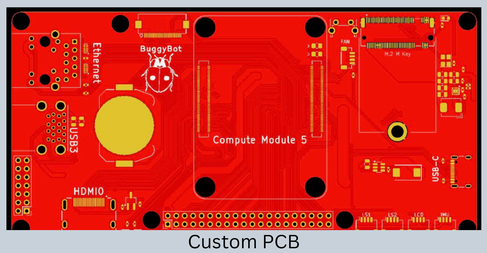
RaspberryPi Camera, Board, and LCD



Camera View and LCD Output of Stop Sign



Camera View and LCD Output of Traffic Signal Ahead Sign



Custom PCB

Conclusion

- Successfully implemented a vision system capable of detecting and classifying multiple traffic sign categories using OpenCV and a trained TensorFlow model.
- Demonstrated real-time traffic sign recognition using live camera footage, displaying predicted classifications and confidence scores.
- Identified challenges: lighting conditions, glare, and false-positive detections, highlighting areas for further optimization.
- Established a functional foundation for integrating traffic sign recognition with the autonomous vehicle's navigation and control systems during Semester 2

Future Work

- Vehicle Assembly:
  - Construct the RC car and integrate motors, sensors onto Custom PCB
- ROS 2 Implementation:
  - Develop communication between the vision system, navigation sensors, and motor control system using ROS 2.
- Traffic Sign Response Testing:
  - Program and evaluate vehicle responses to detected traffic signs, including stopping and adjusting speed.
- Autonomous Navigation Testing:
  - Implement and refine line-following sensors to enable autonomous lane navigation

References: [1] OpenCV, “OpenCV Documentation.” [Online]. Available: https://docs.opencv.org/4.x/ [2] M. Abadi et al., “TensorFlow: Large-scale machine learning on heterogeneous systems,” 2015. [Online]. Available: https://www.tensorflow.org/ [3] Raspberry Pi Ltd., “Raspberry Pi Documentation.” [Online]. Available: https://www.raspberrypi.com/documentation/ [4] KiCad Developers, “KiCad EDA Documentation.” [Online]. Available: https://docs.kicad.org/